

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

AUTOMATED INVENTORY MANAGEMENT SYSTEM FOR PHARMACY AND CONVENIENCE STORE

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The retail pharmacy and convenience store sector plays an important role in providing medicines, healthcare products, and daily necessities to the local community. These establishments are expected to maintain accurate inventory records and efficient customer service to meet the growing demands of consumers. However, many pharmacies and convenience stores still rely on manual methods for inventory tracking and sales recording, which often result in delays, inaccurate records, and operational inefficiencies. Manual processes such as handwritten inventory logs and manual stock counting are time-consuming and prone to human error, leading to inventory discrepancies, stock shortages, overstocking, and customer dissatisfaction.

In addition, outdated scanning systems and manual transaction processing can slow down operations and affect the overall customer experience. Employees may also experience difficulty in monitoring stock levels and updating inventory records in real time, which can reduce productivity and business efficiency. As businesses continue to grow, traditional inventory management methods become less effective in handling increasing numbers of products and transactions.

To address these challenges, this capstone project proposes the development of an Automated Inventory Management System (AIMS) for pharmacies and convenience stores. The proposed system aims to streamline inventory monitoring, automate sales recording, and provide real-time stock updates to improve operational efficiency and accuracy. Through automation, the system is expected to reduce manual errors, improve transaction speed, and enhance customer service. Additionally, the proposed system will help store owners and employees manage inventory more effectively while supporting better decision-making and overall business performance.

1.2 Statement of the Problem

Despite the essential services provided by pharmacies and convenience stores, several operational issues still persist. Many establishments continue to rely on manual processes in managing inventory and sales transactions, resulting in inefficiencies, inaccurate records, and delays in daily operations. In addition, outdated scanning systems contribute to slow transaction processing and customer dissatisfaction.

To address these concerns, this study aims to develop an Automated Inventory Management System (AIMS) for pharmacies and convenience stores.

Specifically, this study seeks to answer the following questions:

1. How can a digital inventory management system be designed and developed to replace manual inventory tracking processes?
2. How can the proposed system improve the accuracy and monitoring of inventory records in pharmacies and convenience stores?
3. How can the system automate sales recording to reduce inventory discrepancies and prevent loss of revenue?
4. How can the integration of an updated barcode scanning system improve transaction speed and customer service efficiency?
5. How can the proposed Automated Inventory Management System improve the overall efficiency and organization of pharmacy and convenience store operations?
6. How acceptable is the proposed Automated Inventory Management System in terms of:
 - functionality,
 - usability,
 - efficiency, and
 - reliability?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop an Automated Inventory Management System (AIMS) that will improve inventory tracking, sales recording, and overall operational efficiency for pharmacies and convenience stores. The proposed system aims to minimize manual processes, reduce human errors, enhance transaction speed, and provide a more organized and reliable method of managing products, sales, and inventory data. Through automation and digital monitoring, the system also seeks to help store staff perform daily operations more efficiently while improving customer service and business productivity.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Analyze the current inventory management practices, user requirements, and existing limitations of manual inventory and sales processes in pharmacies and convenience stores.
2. Design and develop an integrated Point of Sale (POS) system that can automatically record sales transactions and update inventory data in real time.
3. Implement a real-time inventory tracking feature that allows staff to monitor product availability, stock movement, and inventory levels accurately.
4. Develop a low-stock alert and notification system to help prevent product shortages and improve inventory replenishment management.
5. Enhance and integrate barcode scanning functionality to reduce transaction delays, improve scanning accuracy, and speed up the checkout process.
6. Create a user-friendly and intuitive interface that enables staff users to easily navigate, manage inventory records, and process transactions efficiently.
7. Improve the accuracy, organization, and security of inventory and sales records through automated digital data management.
8. Reduce human errors commonly associated with manual inventory counting, handwritten sales recording, and outdated monitoring systems.
9. Evaluate the proposed Automated Inventory Management System in terms of functionality, usability, efficiency, reliability, and overall performance based on user feedback and system testing.

1.4 Scope and Delimitation

This study focuses on the design and development of an Automated Inventory Management System (AIMS) intended for pharmacies and convenience stores. The proposed system aims to improve inventory monitoring, sales recording, and transaction processing through automation and digital data management. The system is designed to assist store owners and staff in managing daily operations more efficiently and accurately.

The scope of the study includes the development of the following features and functionalities:

- Automated Inventory Tracking – The system will automatically monitor product quantities, stock movement, and inventory updates whenever products are added, sold, or restocked.
- Integrated Point of Sale (POS) System – The system will include a POS feature that records sales transactions automatically and updates inventory data in real time.
- Real-Time Stock Updates – The inventory database will provide updated stock information to help staff monitor product availability and prevent stock discrepancies.
- Low Stock Alert Notifications – The system will notify users when product quantities reach a minimum threshold to support timely restocking and inventory management.
- Barcode Scanning Integration – The system will support barcode scanning to improve transaction speed, reduce manual encoding, and enhance accuracy during checkout operations.
- User Management Dashboard – The system will include a dashboard for authorized users to manage products, monitor inventory, access sales records, and oversee system activities.
- Sales and Inventory Reports – The system will generate basic reports related to sales transactions, inventory levels, and product movement for monitoring and documentation purposes.
- User-Friendly Interface – The proposed system will provide an intuitive and easy-to-use interface suitable for pharmacy and convenience store staff with basic computer knowledge.

The study is limited only to the development and testing of a prototype Automated Inventory Management System. The project will focus primarily on inventory and sales management functionalities and will not include advanced features such as:

- predictive analytics,
- artificial intelligence–based forecasting,
- online ordering systems,
- multi-branch synchronization,
- customer loyalty programs,
- accounting integration, and
- large-scale cloud deployment.

In addition, the study will only evaluate the system in terms of functionality, usability, efficiency, and reliability within the selected testing environment. The system is intended for academic and prototype purposes only and is not designed for full commercial implementation at this stage.

1.5 Significance of the Study

The proposed Automated Inventory Management System (AIMS) is expected to provide significant benefits to pharmacies and convenience stores by improving inventory management, sales recording, and transaction processing. Through automation and real-time monitoring, the system aims to reduce manual work, minimize errors, and enhance the overall efficiency of store operations.

The study will benefit the following:

Users

The system will benefit users, particularly customers and store staff, by providing more accurate inventory tracking and faster transaction processing. Customers will experience improved service due to reduced waiting time during purchases and better product availability monitoring. Staff users will also benefit from a more organized and efficient method of recording sales, checking stock levels, and managing inventory information.

Store Owners

Store owners will benefit from the proposed system through improved operational efficiency and better inventory management. The automation of inventory tracking and sales recording can help reduce human errors, prevent inventory discrepancies, and minimize financial losses caused by inaccurate stock monitoring. The system can also assist owners in making informed decisions by providing updated inventory and sales records.

Employees and Staff

The proposed system will help employees and staff perform their daily tasks more efficiently. By automating manual processes such as stock counting, sales recording, and barcode scanning, the system can reduce workload, improve productivity, and simplify transaction processing. The user-friendly interface will also make the system easier to learn and operate.

BSIT Students

This study will serve as a practical application of the knowledge and skills acquired by Bachelor of Science in Information Technology (BSIT) students in the fields of software development, database management, system analysis, and system integration. It will also provide students with experience in designing and developing real-world information systems that address actual business problems.

Future Researchers

Future researchers may use this study as a reference for further research related to automated inventory systems, retail management systems, and Point of Sale (POS) technologies. The findings, methodologies, and system design presented in this study may help future studies improve and expand similar systems by incorporating additional features and technologies.

Pharmacies and Convenience Stores

Pharmacies and convenience stores may benefit from this study by gaining a more modern and efficient approach to inventory and sales management. The implementation of automated systems can improve accuracy, organization, and overall business operations, helping establishments provide better service to customers and maintain more reliable inventory records.

1.6 Definition of Terms

Automated Inventory Management System (AIMS)

An Automated Inventory Management System is a computerized system designed to monitor, organize, and manage inventory levels, product movement, and sales transactions automatically. In this study, the system is used to reduce manual inventory processes, improve accuracy, and enhance operational efficiency in pharmacies and convenience stores.

Inventory Tracking

Inventory tracking refers to the process of monitoring the quantity, availability, and movement of products within a store or business establishment. This includes recording incoming stocks, sold items, and remaining inventory levels to maintain accurate product records.

Point of Sale (POS)

Point of Sale (POS) refers to a computerized system used to process customer purchases and sales transactions. In this study, the POS system automatically records sales, updates inventory levels, and helps improve transaction speed and accuracy.

Real-Time Updates

Real-time updates refer to the immediate and automatic updating of inventory and sales data whenever transactions or stock movements occur. This feature allows users to access current and accurate information without delays.

Barcode Scanning System

A barcode scanning system is a technology used to read product barcodes for faster and more accurate item identification during transactions and inventory management. In this study, barcode scanning helps reduce manual encoding and speeds up the checkout process.

Sales Recording

Sales recording refers to the process of documenting completed sales transactions, including product details, quantities sold, prices, and payment information. The proposed system automates this process to improve accuracy and organization.

Low Stock Alert

A low stock alert is a system-generated notification that informs users when the quantity of a product reaches a minimum or critical level. This feature helps prevent stock shortages and supports timely restocking of products.

User Management

User management refers to the process of controlling and organizing user access within the system. It includes managing user accounts, roles, permissions, and login credentials to ensure proper system security and authorized access.

Database

A database is an organized collection of digital information stored electronically for easy access, management, and updating. In this study, the database stores inventory records, sales transactions, product details, and user information.

Operational Efficiency

Operational efficiency refers to the ability of a business or system to perform tasks accurately, quickly, and with minimal wasted time, effort, or resources. The proposed system aims to improve operational efficiency by automating inventory and sales management processes.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Foreign studies show that automated inventory management systems and web-based Point of Sale (POS) technologies are effective in improving inventory monitoring, sales recording, transaction processing, and operational efficiency in retail businesses, pharmacies, and convenience stores. These studies are relevant to the present project because they demonstrate how manual inventory and sales processes can be improved through automation and digital systems.

2.1.1

A foreign study conducted by Saleem (2020) examined the impact of automated inventory management systems on supply chain risk management and organizational performance among manufacturing firms in Pakistan. The study focused on determining how automation technologies improve inventory monitoring, operational efficiency, and business productivity. The researchers explained that many organizations still rely on traditional inventory management methods, which often result in inaccurate inventory records, delayed transactions, product shortages, overstocking, and poor coordination between departments. These problems negatively affect supply chain performance and reduce the overall efficiency of business operations.

The study emphasized that automated inventory management systems are important because they help organizations manage inventory more effectively through computerized processes and digital technologies. The researchers specifically examined several technologies commonly used in automated inventory systems, including Radio Frequency Identification (RFID), Enterprise Resource Planning (ERP), Electronic Data Interchange (EDI), and Material Requirements Planning (MRP). According to the study, these technologies improve the monitoring, recording, and movement of inventory within organizations while reducing manual work and operational delays..

The findings of the study revealed that automated inventory management systems significantly improved inventory accuracy and operational efficiency within manufacturing firms. Through automation, organizations were able to track inventory levels more accurately and update inventory information in real time. This reduced the occurrence of inventory discrepancies caused by manual recording and minimized errors in stock monitoring. The researchers explained that real-time inventory updates enabled organizations to respond more quickly to inventory shortages and operational problems, helping improve workflow and supply chain coordination.

Furthermore, the study found that RFID technology improved product tracking and inventory visibility throughout the supply chain. By using RFID tags and scanning technologies, companies were able to monitor product movement more efficiently and reduce problems such as misplaced inventory and delivery errors. The study also highlighted that ERP systems improved communication and information sharing among departments by centralizing inventory data into a unified digital platform. Because information could be accessed quickly and accurately, managers were able to make better decisions related to inventory replenishment, purchasing, and stock management.

The research also explained that Electronic Data Interchange (EDI) technologies improved communication between supply chain partners and helped organizations exchange business information electronically. This reduced delays associated with manual paperwork and improved transaction speed. In addition, Material Requirements Planning (MRP) systems improved production planning and inventory scheduling by helping companies determine the quantity of materials needed for operations. These technologies collectively improved organizational productivity and reduced operational risks associated with inventory management.

Another important finding of the study was the positive effect of automation on customer service and organizational performance. Because organizations were able to monitor inventory levels more effectively, products became more readily available and delivery delays were reduced. This improved customer satisfaction and strengthened business performance. The researchers emphasized that automation not only improves inventory accuracy but also enhances the ability of organizations to provide reliable and efficient services to customers.

The study further highlighted the importance of employee training and development in the successful implementation of automated inventory management systems. According to the researchers, organizations that invested in employee training achieved better results because employees became more capable of using automated technologies efficiently. The study noted that some employees initially experienced difficulties adapting to automated systems due to limited technical knowledge and resistance to organizational change. However, proper training programs helped employees understand the system more effectively and increased their confidence in using digital technologies.

Despite the positive findings, the study identified several challenges encountered during the implementation of automated inventory systems. The transition from manual inventory management to computerized systems required financial investment, technical infrastructure, and skilled personnel. Organizations needed the support of IT professionals and technology specialists to install, configure, and maintain the systems properly. In addition, the implementation process required time and organizational adjustments because employees had to adapt to new operational procedures and digital workflows.

2.1.2

A foreign study conducted by Akinsanya (2019) focused on the design and implementation of a computerized stock management system for MIDE Supermarket. The study aimed to address the operational problems associated with manual inventory monitoring and stock recording within supermarket businesses. According to the researchers, many retail establishments continue to use traditional methods of recording inventory, which often lead to inaccurate stock information, slow transaction processing, loss of records, and difficulties in monitoring product movement. These challenges negatively affect business productivity and may reduce customer satisfaction due to delays and inventory inconsistencies.

The study explained that manual inventory management consumes a significant amount of time and effort because employees are required to record stock information manually and regularly check inventory levels using paper-based records. The researchers noted that this process increases the possibility of human error, including incorrect stock counting, duplicated entries, and missing records. As inventory volume increases, businesses may struggle to maintain accurate records, which can lead to overstocking or shortages of important products.

To solve these problems, the researchers developed a computerized stock management system designed to automate inventory monitoring and product management within the supermarket. The proposed system allowed employees to record inventory data digitally, monitor stock levels in real time, and update inventory records automatically whenever products were added or sold. The system also improved the organization of inventory data by storing product information in a centralized database that could easily be accessed and updated by authorized personnel.

The findings of the study revealed that the implementation of the computerized stock management system significantly improved inventory accuracy and operational efficiency within the supermarket. The automated system reduced manual work associated with stock recording and minimized human errors caused by handwritten inventory records. Employees were able to monitor inventory levels more efficiently because the system provided updated product information whenever transactions occurred. This improved the reliability of inventory records and reduced problems associated with inventory discrepancies.

Furthermore, the study highlighted that the computerized system improved transaction processing and customer service within the supermarket. Because inventory information could be accessed quickly, employees were able to identify product availability more efficiently and assist customers faster. The researchers explained that automation also improved the speed of inventory-related operations, reducing delays caused by manual checking and paperwork. As a result, customer satisfaction improved because products were more readily available and transactions became more organized.

Another important contribution of the study was the improvement of inventory security and record management. The computerized system allowed businesses to maintain digital copies of inventory data, reducing the risks associated with lost or damaged paper records. Managers were also able to generate reports related to stock movement, inventory levels, and sales performance more easily. These reports helped management make informed decisions regarding inventory replenishment and product purchasing.

Despite the positive outcomes, the study identified several challenges encountered during system implementation. The supermarket needed financial resources to acquire computer equipment and install the software system. Employees also required training to understand how to operate the computerized inventory system properly. Some workers initially experienced difficulty adjusting to the new system because they were more familiar with traditional inventory methods. However, continuous training and practice helped employees adapt to the digital process over time.

The study concluded that computerized inventory management systems improve operational efficiency, inventory accuracy, and customer service within retail establishments. The researchers emphasized that although implementing automated systems requires investment and technical support, the long-term benefits include reduced human error, improved inventory monitoring, faster transaction processing, and better organizational performance.

This foreign study is relevant to the present study because it demonstrates how computerized inventory systems improve inventory monitoring and operational efficiency in retail businesses. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of digital technologies to automate inventory tracking, reduce manual work, and improve sales recording processes. The findings also highlight the importance of real-time inventory updates and employee adaptation in successfully implementing inventory management systems.

2.1.3

A foreign study conducted by Abisoye, Fatoba, and Blessing (2013) examined the design of computerized inventory management systems for supermarkets and their impact on business operations. The study aimed to determine how automation technologies improve inventory control, stock monitoring, and organizational productivity within retail establishments. The researchers explained that supermarkets commonly experience operational problems related to manual inventory management, including inaccurate stock records, delayed inventory updates, poor product monitoring, and inefficient transaction processing. These problems often affect customer service and increase operational costs.

The study emphasized that inventory management plays an important role in the success of retail businesses because it helps organizations monitor product movement, maintain sufficient stock levels, and avoid inventory shortages. According to the researchers, businesses that rely on manual inventory processes often encounter difficulties in maintaining accurate inventory records due to the large volume of products handled daily. This increases the risk of inventory discrepancies, misplaced products, and poor decision-making regarding stock replenishment.

To address these challenges, the researchers designed a computerized inventory management system capable of automating inventory tracking and stock recording processes. The system provided features that allowed users to store product information digitally, monitor inventory levels, and process inventory updates automatically whenever products were purchased or restocked. The researchers explained that the system also improved inventory organization by centralizing product data within a computerized database.

The findings of the study revealed that the computerized inventory management system significantly improved inventory accuracy and operational efficiency within supermarkets. Through automation, employees were able to access updated inventory information more quickly and reduce the occurrence of manual recording errors. The researchers noted that the system improved stock monitoring because product quantities were updated automatically in real time. This allowed businesses to identify low-stock items more effectively and prevent product shortages.

Furthermore, the study found that the computerized system improved transaction speed and workflow efficiency within supermarkets. Employees no longer needed to manually verify stock records because the system automatically processed inventory updates and sales information. This reduced delays during transactions and improved the overall customer experience. The researchers also explained that automation reduced employee workload by simplifying inventory-related tasks and minimizing repetitive manual procedures.

Another important finding of the study was the improvement of business decision-making through digital reporting and inventory analysis. Managers were able to generate inventory reports more efficiently and analyze product movement using the computerized system. This helped businesses identify fast-moving and slow-moving products, improve purchasing decisions, and maintain balanced inventory levels. The researchers emphasized that accurate inventory information contributes to better organizational planning and operational productivity.

The study also discussed several challenges encountered during system implementation. Businesses needed financial resources to purchase computer equipment and software tools necessary for the computerized inventory system. Employees also required technical training to become familiar with the new digital processes. Some workers initially resisted the transition because they were accustomed to traditional inventory methods. However, the researchers noted that continuous system usage and proper training improved employee confidence and system acceptance over time.

The study concluded that computerized inventory management systems improve inventory control, transaction efficiency, organizational productivity, and customer service within supermarkets. The researchers emphasized that automation provides businesses with more accurate inventory information and simplifies inventory-related operations. Although implementation may involve financial costs and technical challenges, the long-term advantages include reduced human errors, improved workflow, and better inventory management.

This foreign study is relevant to the present study because it demonstrates how computerized inventory systems improve inventory accuracy, transaction processing, and operational efficiency in retail businesses. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of automation technologies to simplify inventory tracking and improve sales recording. The findings also support the integration of digital databases and real-time inventory monitoring in modern inventory management systems.

2.1.4

A foreign study conducted by Samuel and Ondiek (2014) investigated the effects of inventory management automation on the performance of supermarkets in Western Kenya. The study aimed to determine how automated inventory systems influence inventory control, operational efficiency, and business performance within retail establishments. The researchers explained that many supermarkets experience operational problems caused by manual inventory management practices, including stock shortages, inaccurate inventory records, delayed stock updates, and inefficient monitoring procedures. These issues often reduce organizational productivity and negatively affect customer satisfaction.

The study emphasized that inventory management automation is important because it improves the ability of businesses to monitor inventory levels and manage stock movement more efficiently. According to the researchers, traditional inventory methods rely heavily on manual work, which increases the risk of human error and delays in processing inventory information. As businesses grow and inventory volume increases, manual processes become less effective and more difficult to manage.

The researchers examined how automated inventory technologies improve retail operations by reducing manual tasks and improving the accuracy of inventory information. The study explained that automated systems allow supermarkets to update inventory records automatically whenever products are sold, restocked, or transferred. This improves inventory visibility and helps businesses maintain updated stock information in real time.

The findings of the study revealed that inventory management automation significantly improved supermarket performance and operational efficiency. Through automated systems, businesses were able to monitor inventory more accurately and reduce problems associated with stock discrepancies. The researchers found that automation improved product availability because inventory levels could be tracked continuously and low-stock products could be identified immediately. This helped supermarkets reduce product shortages and improve customer service.

Furthermore, the study highlighted that automation improved transaction processing and inventory organization within supermarkets. Employees were able to complete inventory-related tasks more efficiently because the system reduced the need for manual recording and repetitive paperwork. The researchers explained that automated systems also improved workflow efficiency by simplifying inventory monitoring procedures and reducing delays associated with manual inventory checking.

Another important finding of the study was the positive effect of automation on decision-making and organizational productivity. Managers were able to access accurate inventory reports and monitor stock movement more effectively using automated systems. This helped businesses improve purchasing decisions, reduce unnecessary inventory costs, and maintain balanced stock levels. The study also noted that automated inventory systems contribute to better warehouse organization and improved operational planning.

Despite the positive findings, the study identified several challenges related to the implementation of automated inventory systems. Supermarkets needed financial investment to acquire technology infrastructure and computerized equipment. Employees also required training programs to understand how to operate the automated systems correctly. Some employees experienced difficulties adapting to new technologies due to limited computer skills and resistance to organizational changes. However, the researchers explained that proper employee support and training improved system acceptance and operational performance.

The study concluded that automated inventory management systems positively affect supermarket performance, inventory control, and operational efficiency. The researchers emphasized that automation helps businesses improve inventory monitoring, reduce operational errors, and enhance customer service. Although implementation may involve technical and financial challenges, the long-term benefits include improved productivity, reduced inventory discrepancies, and better business operations.

This foreign study is relevant to the present study because it demonstrates how inventory automation improves operational efficiency, inventory monitoring, and business performance within retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of automated technologies to improve inventory accuracy, reduce manual work, and enhance sales and inventory management processes. The findings also strengthen the importance of real-time inventory updates and digital inventory monitoring in modern retail operations.

2.1.5

A foreign study conducted by Koumanakos (2008) examined the relationship between inventory management practices and business performance among manufacturing organizations in Greece. The study aimed to determine how efficient inventory management systems contribute to organizational productivity, profitability, and operational efficiency. The researcher explained that many businesses experience difficulties in maintaining accurate inventory records because traditional inventory methods rely heavily on manual monitoring and paperwork. These manual processes often result in inventory shortages, overstocking, delayed transactions, and increased operational costs, which negatively affect business performance.

The findings of the study revealed that organizations with efficient inventory management systems experienced improved operational productivity and better financial performance. The study emphasized that businesses using automated inventory monitoring methods were more capable of controlling stock levels, reducing unnecessary storage costs, and maintaining accurate inventory records. Through automation and computerized inventory tracking, organizations were able to improve workflow efficiency and reduce the occurrence of inventory discrepancies caused by human error.

Furthermore, the study highlighted that inventory management systems help businesses improve decision-making by providing accurate and updated inventory information. Managers were able to identify fast-moving and slow-moving products more effectively and make informed purchasing decisions. This improved inventory planning and reduced losses associated with poor stock management. The researcher also noted that organizations with improved inventory systems were more capable of responding to customer demands and maintaining product availability.

Despite the positive findings, the study identified challenges related to implementing computerized inventory systems. Organizations needed financial investment for technology infrastructure and employee training. Some employees initially struggled to adapt to automated technologies due to limited technical knowledge. However, proper training and continuous system usage improved employee acceptance and operational performance over time.

The study concluded that effective inventory management systems significantly improve organizational efficiency, inventory accuracy, and financial performance. The researcher emphasized that automation plays an important role in reducing operational problems and improving business productivity.

This study is relevant to the present research because it supports the use of automated inventory management systems in improving inventory monitoring, operational efficiency, and decision-making processes. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study highlights the importance of accurate inventory tracking and digital inventory management.

2.1.6

A foreign study conducted by Prempeh (2015) investigated inventory management practices and their effect on operational performance in retail businesses in Ghana. The study aimed to examine how computerized inventory systems improve inventory control, reduce operational costs, and enhance business productivity. According to the researcher, retail businesses that rely on manual inventory methods commonly experience inventory inaccuracies, stock shortages, and delays in transaction processing.

The findings revealed that automated inventory systems improved inventory accuracy and reduced losses associated with poor stock monitoring. Businesses using computerized inventory technologies were able to maintain updated inventory records and monitor stock movement more effectively. The study also showed that automated systems improved customer satisfaction because products became more readily available and transactions were processed more efficiently.

The research further explained that inventory automation improved organizational productivity by reducing repetitive manual tasks and simplifying inventory-related operations. Managers were able to generate reports and monitor inventory data in real time, allowing them to make better inventory decisions and improve stock replenishment planning.

Despite these advantages, the study identified implementation challenges such as financial costs, technical maintenance, and employee resistance to new technologies. However, the researcher emphasized that the long-term benefits of automation outweigh the initial challenges because automated inventory systems improve operational efficiency and business performance.

This study is relevant to the present study because it demonstrates how inventory automation improves inventory monitoring, transaction processing, and operational efficiency in retail establishments. It also supports the integration of real-time inventory tracking and computerized inventory systems in pharmacies and convenience stores.

2.1.7

A foreign study conducted by Rajeev (2010) explored the importance of inventory management in improving organizational efficiency within manufacturing and retail businesses. The study emphasized that inventory management is essential because it affects product availability, customer satisfaction, and operational productivity. The researcher explained that manual inventory methods often lead to inaccurate stock records and delays in inventory updates.

The findings revealed that businesses using computerized inventory systems experienced improved inventory control and better operational performance. Through automated monitoring and digital record management, organizations were able to reduce inventory discrepancies and improve workflow efficiency. The study also highlighted that inventory automation reduced unnecessary storage costs and improved product availability.

Furthermore, the study explained that inventory management systems contribute to better organizational planning because managers can access updated inventory information quickly. This improved decision-making processes and reduced losses associated with poor inventory monitoring.

The study concluded that inventory automation is important for improving organizational productivity and customer service. It emphasized that businesses adopting digital inventory systems gain advantages in inventory accuracy, operational efficiency, and inventory planning.

This study is relevant to the present research because it supports the importance of automated inventory systems in improving business operations and inventory monitoring within pharmacies and convenience stores.

2.1.8

A foreign study conducted by Wild (2017) examined the role of computerized inventory systems in modern retail and warehouse operations. The study aimed to determine how automation technologies improve inventory visibility, stock control, and customer satisfaction. The researcher explained that businesses using manual inventory systems often encounter inventory inaccuracies and operational inefficiencies.

The findings showed that automated inventory systems improved stock monitoring and reduced inventory discrepancies. Through digital inventory databases and barcode technologies, businesses were able to process inventory transactions faster and maintain accurate inventory records. The study also noted that automation reduced delays in inventory updates and improved operational workflow.

Another important finding was that inventory automation improved customer service because businesses could provide accurate product availability information and faster transaction processing. Managers also benefited from real-time inventory reports and better inventory planning.

The study concluded that computerized inventory systems improve organizational productivity and inventory efficiency. It emphasized that businesses adopting automated inventory technologies are better equipped to manage inventory-related operations and customer demands.

This study is relevant to the current study because it highlights the importance of barcode integration, real-time updates, and computerized inventory tracking in improving retail operations.

2.1.9

A foreign study conducted by Stevenson (2018) focused on the importance of inventory management systems in retail and supply chain operations. The study aimed to determine how automation technologies improve stock control and reduce operational risks. According to the researcher, traditional inventory methods increase the risk of stock shortages, delayed transactions, and inaccurate inventory data.

The findings revealed that automated inventory systems improved inventory visibility and operational coordination. Businesses using computerized systems were able to monitor inventory levels more effectively and reduce operational delays. The study also emphasized that barcode scanning technologies improved transaction speed and minimized human error.

Furthermore, the study explained that automated systems improved organizational decision-making by providing accurate inventory reports and sales information. Managers could analyze inventory movement more efficiently and improve stock replenishment decisions.

The study concluded that automation contributes significantly to operational efficiency and customer satisfaction in retail businesses. Businesses using automated systems experienced improved workflow, inventory accuracy, and transaction processing.

This study is relevant to the present study because it supports the integration of barcode scanning, real-time inventory monitoring, and automated sales recording within pharmacies and convenience stores.

2.1.10

A foreign study conducted on Alfa Company, a medium-sized distributor of laboratory equipment and supplies in Iraq, examined the implementation of an automated inventory management system and its effects on the company's business operations. The study aimed to determine how automation could improve inventory control, operational efficiency, and customer service in small and medium-sized enterprises (SMEs). The researchers emphasized that traditional inventory management methods often lead to problems such as inaccurate stock records, delayed transactions, overstocking, and product shortages, which may negatively affect the overall performance of a business.

The findings of the study revealed that the implementation of the automated inventory management system significantly improved the company's ability to monitor and manage inventory levels. Through the use of computerized processes, the company was able to track products more accurately, reduce human errors, and maintain updated inventory records in real time. The system also helped minimize storage expenses by preventing excessive stock accumulation and improving warehouse organization. In addition, the study noted that customer satisfaction improved because the company was able to provide faster and more reliable services, ensuring that needed products were readily available and delivered efficiently.

Furthermore, the research highlighted that automated inventory systems contribute to better decision-making within organizations. Because inventory data could be accessed quickly and accurately, managers were able to analyze stock movement, identify fast-moving and slow-moving items, and make informed purchasing decisions. This improved the company's operational productivity and reduced unnecessary costs associated with inventory mismanagement. The study also emphasized that automation supports business growth by enabling companies to handle larger volumes of inventory and transactions more effectively.

Despite the positive outcomes, the study identified several challenges encountered during the implementation process. The transition from manual to automated inventory management required a significant investment of time, financial resources, and technical expertise. The company needed the assistance of IT professionals to install, configure, and maintain the system properly. Additionally, employees had to undergo training programs to familiarize themselves with the new technology and operational procedures. Some workers initially experienced difficulty adapting to the system due to limited technical knowledge and resistance to organizational change.

The study further explained that collaboration between businesses, IT consultants, and software suppliers played an important role in the successful implementation of the automated inventory management system. These professionals assisted the company in evaluating the advantages and costs of automation while ensuring that the system met the specific needs of the organization. The researchers concluded that although implementing an automated inventory system may involve challenges, the long-term benefits outweigh the difficulties because it enhances efficiency, accuracy, productivity, and customer service.

This foreign study is relevant to the present study because it demonstrates how automated inventory management systems can improve inventory accuracy, operational efficiency, and business performance. It also supports the idea that adopting computerized systems can reduce manual errors and simplify inventory-related tasks. The findings serve as a valuable reference for the current research, particularly in understanding the importance of automation in inventory management and its impact on organizational effectiveness.

2.1.11

Researchers from Jordan conducted a study that investigated the impact of a warehouse management system on supply chain performance, particularly in improving inventory management efficiency and reducing operational effort within the telecommunications sector. The study focused on the warehouse operations of a leading telecommunications service provider and aimed to determine how a customized warehouse management system could enhance workflow efficiency, inventory accuracy, and supply chain reliability. The researchers examined existing warehouse procedures before developing and implementing a software-based inventory management system capable of handling warehouse transactions and operational processes more effectively.

The study explained that traditional manual inventory systems often create operational inefficiencies due to slow processing, inaccurate inventory records, and delays in handling products and transactions. These problems may negatively affect supply chain performance and reduce the ability of companies to respond efficiently to customer demands. To address these issues, the researchers customized a warehouse management software system designed to automate and simplify inventory-related operations. The system was tested to evaluate its effectiveness in improving warehouse workflow, handling inventory transactions in a timely manner, and supporting overall supply chain operations.

The findings of the study revealed that the implementation of the warehouse management system significantly improved operational efficiency and inventory management processes. The system reduced the amount of manual labor required in handling inventory transactions, resulting in faster and more reliable warehouse operations. Through automation, inventory data became more accurate and accessible, which helped employees monitor stock levels effectively and minimize errors in inventory recording and product handling. The researchers also noted that the software improved communication and coordination among warehouse personnel, contributing to smoother and more organized operations.

In addition, the study examined the physical layout of the warehouse and introduced a production station within the facility to optimize warehouse space utilization. The production station involved three important processes: bundling, labeling, and repackaging of products. According to the researchers, integrating these production activities within the warehouse improved workflow organization and allowed the company to maximize available storage and operational space. This adjustment contributed to better productivity and more efficient handling of products throughout the warehouse operations.

Furthermore, the warehouse management system handled three major phases of the product lifecycle, namely receiving, processing, and distribution of SIM cards and prepaid scratch cards. The researchers discussed each phase in detail and identified operational gaps and inefficiencies in the previous manual procedures. By automating these phases, the company improved the speed and accuracy of product movement within the supply chain. The study also highlighted that the customized software system helped reduce delays, improve transaction monitoring, and support more efficient distribution processes.

The researchers concluded that warehouse management systems play a vital role in strengthening supply chain performance and improving inventory management in the telecommunications industry. The study emphasized that automated inventory systems are more efficient and reliable than traditional manual methods because they reduce operational effort, improve inventory visibility, and support faster decision-making. Moreover, the researchers stated that the study may serve as a practical guide for organizations and future researchers interested in comparing automated and manual inventory management systems. The findings also highlighted the importance of developing customized systems that can address operational challenges and reduce supply chain disruptions.

This study is relevant to the present research because it demonstrates how automated warehouse and inventory management systems improve operational efficiency, workflow organization, and inventory accuracy. It supports the idea that computerized inventory systems can simplify business processes, reduce manual errors, and enhance supply chain performance. The study also provides valuable information regarding the importance of automation in inventory and warehouse operations, which may contribute to the development of more efficient inventory management systems in the current research.

2.1.12

A foreign study conducted by Gunasekaran and Ngai (2004) examined the role of information systems and automation technologies in supply chain and inventory management within business organizations. The study aimed to determine how computerized inventory systems improve operational efficiency, inventory visibility, organizational productivity, and overall business performance. The researchers explained that traditional inventory management methods are often ineffective because they rely heavily on manual processes, paperwork, and delayed communication between departments. These problems may result in inaccurate inventory records, stock shortages, delayed transactions, poor inventory coordination, and customer dissatisfaction.

The researchers emphasized that inventory management is one of the most important operational functions within organizations because inventory directly affects customer service, productivity, profitability, and business continuity. According to the study, businesses that fail to maintain accurate inventory records often encounter financial losses due to overstocking, product expiration, theft, and poor inventory planning. The researchers also noted that manual inventory systems consume significant time and labor because employees are required to encode information manually, conduct physical inventory checking repeatedly, and process inventory reports through traditional paperwork.

To address these operational challenges, the study explored the use of computerized information systems and digital inventory technologies in improving inventory and supply chain management. The researchers highlighted that automation technologies allow organizations to store, process, and retrieve inventory information more efficiently through centralized digital databases. The study also explained that automated inventory systems provide real-time inventory updates, enabling businesses to monitor stock movement immediately whenever products are purchased, transferred, or restocked.

The findings of the study revealed that organizations using automated inventory systems experienced significant improvements in inventory monitoring and operational efficiency. Through automation, businesses were able to reduce human errors associated with manual inventory recording and improve the accuracy of inventory information. The researchers explained that real-time inventory monitoring allowed organizations to identify stock shortages quickly, prevent overstocking, and maintain balanced inventory levels. As a result, businesses were able to improve inventory coordination and reduce operational delays.

Furthermore, the study found that computerized inventory systems improved communication and information sharing within organizations. Inventory information could be accessed by multiple departments simultaneously, improving coordination between purchasing, sales, warehousing, and management personnel. This reduced delays caused by manual reporting and improved organizational workflow. Managers were also able to generate reports more efficiently and analyze inventory data to support better operational planning and decision-making.

Another important finding of the study was the positive effect of automation on customer service and organizational productivity. Businesses using automated systems were able to process transactions faster, provide more accurate product availability information, and respond to customer demands more efficiently. The researchers emphasized that customers benefit from automation because businesses can maintain adequate stock levels and reduce delays in service delivery. In addition, automation reduced repetitive manual tasks, allowing employees to focus on more productive operational activities.

The study also highlighted several technologies commonly used in automated inventory management systems, including barcode scanning systems, enterprise resource planning systems, and electronic data interchange technologies. According to the researchers, barcode technologies improve transaction speed and inventory accuracy by reducing manual encoding errors. Enterprise systems help organizations centralize inventory information and improve communication across departments, while electronic data exchange technologies improve coordination with suppliers and business partners.

Despite the positive outcomes of automation, the study identified several challenges associated with implementing computerized inventory systems. Organizations needed financial investment to acquire hardware, software, and technical infrastructure necessary for automation. Businesses also required technical experts to install, configure, and maintain the systems properly. The researchers explained that employees may initially experience difficulty adapting to automated technologies due to limited computer knowledge and resistance to organizational changes. However, proper training and continuous system use improved employee confidence and system acceptance over time.

The study further explained that organizational support and management commitment are important factors in the successful implementation of automated inventory systems. Managers must provide employees with adequate training, technical support, and guidance to ensure effective system adoption. The researchers also emphasized that collaboration between technology providers and organizations helps ensure that inventory systems meet operational requirements and business objectives.

The study concluded that computerized inventory systems and automation technologies significantly improve inventory management, organizational coordination, operational productivity, and customer service. The researchers emphasized that although implementing automated inventory systems may involve technical and financial challenges, the long-term benefits outweigh the difficulties because automation improves inventory accuracy, simplifies operational processes, reduces delays, and enhances overall organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems improve inventory monitoring, transaction processing, and operational efficiency within organizations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of digital technologies, barcode scanning systems, and real-time inventory monitoring to reduce manual errors and improve inventory-related operations. The findings also strengthen the importance of automation in improving customer service, workflow efficiency, and organizational productivity.

2.1.13

A foreign study conducted by DeHoratius and Raman (2008) investigated inventory record inaccuracies in retail establishments and examined how automated inventory systems improve inventory control and operational efficiency. The study aimed to identify the common causes of inventory discrepancies in retail businesses and determine how technology-based inventory systems reduce errors associated with manual inventory management. The researchers explained that accurate inventory records are important because they directly affect sales performance, customer satisfaction, operational productivity, and organizational profitability. However, many retail businesses continue to experience inventory-related problems due to outdated manual inventory practices and inefficient monitoring procedures.

The researchers emphasized that inventory inaccuracies are among the most common operational problems in retail businesses. According to the study, inventory discrepancies occur when the recorded quantity of products does not match the actual physical inventory available in the store or warehouse. These discrepancies are often caused by manual recording errors, theft, misplaced items, delayed inventory updates, and poor inventory coordination. The study noted that inventory inaccuracies may result in product shortages, delayed customer service, lost sales opportunities, and poor business decision-making.

To better understand the problem, the researchers analyzed inventory management practices in retail establishments and examined how businesses monitor and update inventory records. The study explained that traditional inventory systems require employees to manually record stock movement and conduct physical inventory counting regularly. This process is time-consuming and increases the likelihood of human error, especially when businesses handle large volumes of products and transactions daily. Because inventory information is not updated immediately, businesses may struggle to maintain accurate inventory records and respond effectively to operational problems.

The findings of the study revealed that businesses using automated inventory systems experienced significant improvements in inventory accuracy and operational performance. Through computerized inventory monitoring and digital record management, organizations were able to reduce errors associated with manual inventory processes. The researchers explained that automated systems update inventory records in real time whenever products are sold, transferred, or restocked. This allowed businesses to monitor stock movement more accurately and identify inventory discrepancies more efficiently.

Furthermore, the study found that barcode scanning technologies played an important role in improving inventory accuracy and transaction efficiency. Barcode systems minimized errors caused by manual product encoding and improved the speed of inventory-related operations. Employees were able to process transactions faster and update inventory information automatically, reducing delays associated with traditional inventory recording methods. The researchers also noted that barcode-integrated systems improved inventory visibility because product information could be accessed quickly and accurately through computerized databases.

Another important finding of the study was the improvement of organizational decision-making through real-time inventory information. Managers were able to generate inventory reports more efficiently and analyze product movement using computerized inventory systems. This helped businesses identify fast-moving and slow-moving products, improve purchasing decisions, and maintain balanced inventory levels. The study emphasized that accurate inventory information supports better inventory planning and reduces losses associated with overstocking and stock shortages.

The study also highlighted the positive effect of automation on customer service and business productivity. Businesses using automated inventory systems were more capable of maintaining product availability and reducing delays during customer transactions. Customers benefited from improved service because businesses could provide accurate information regarding product availability and process transactions more efficiently. In addition, automation reduced repetitive manual tasks, allowing employees to focus on more productive operational activities.

Despite the positive findings, the study identified several challenges associated with implementing automated inventory systems. Businesses required financial investment for technology infrastructure, barcode equipment, and computerized software systems. Employees also needed technical training to understand how to operate the systems properly. Some workers initially experienced difficulties adapting to new technologies because they were more familiar with manual inventory procedures. However, continuous training and regular system use improved employee confidence and system acceptance over time.

The researchers concluded that automated inventory systems significantly improve inventory accuracy, operational efficiency, transaction processing, and organizational productivity within retail businesses. The study emphasized that although implementing computerized inventory systems may involve technical and financial challenges, the long-term benefits include reduced inventory discrepancies, improved workflow management, enhanced customer service, and better inventory planning.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems and barcode technologies improve inventory monitoring and transaction efficiency within retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of real-time inventory monitoring, barcode scanning, and computerized inventory management to reduce manual errors and improve operational performance.

2.1.14

A foreign study conducted by Sharma, Garg, and Agarwal (2011) examined the implementation of barcode technology in inventory management systems and its effect on retail operations and organizational efficiency. The study aimed to determine how barcode scanning technologies improve transaction processing, inventory accuracy, product monitoring, and customer service in retail establishments. The researchers explained that many businesses continue to experience operational inefficiencies due to manual inventory recording and traditional checkout procedures. These manual methods often result in delayed transactions, incorrect inventory records, inventory discrepancies, and customer dissatisfaction.

The researchers emphasized that barcode technology has become an important component of modern inventory management because it improves the speed and accuracy of inventory-related operations. According to the study, traditional inventory systems rely heavily on manual encoding and handwritten records, which increase the likelihood of human error and delay operational processes. Employees may accidentally input incorrect product information or fail to update inventory records immediately, leading to inaccurate stock data and operational inefficiencies.

The study explored how barcode scanning systems function within inventory and retail operations. Barcode technology allows products to be identified quickly through machine-readable codes attached to product packaging. When scanned using barcode readers, product information is automatically retrieved from the inventory database and updated in real time. The researchers explained that this automation process reduces manual work and improves inventory monitoring efficiency.

The findings of the study revealed that businesses using barcode-integrated inventory systems experienced significant improvements in transaction speed and inventory accuracy. Barcode scanning reduced delays during checkout operations because employees no longer needed to encode product information manually. The automated process minimized typing errors and improved the accuracy of sales recording and inventory updates. As a result, businesses were able to maintain more reliable inventory records and reduce problems associated with inventory discrepancies.

Furthermore, the study found that barcode systems improved inventory visibility and product tracking within organizations. Inventory information could be updated automatically whenever products were sold, transferred, or restocked. This enabled businesses to monitor inventory movement more efficiently and identify low-stock products quickly. Managers were also able to access inventory reports more accurately and improve purchasing and inventory planning decisions.

Another important finding of the study was the improvement of customer satisfaction and operational productivity. Businesses using barcode systems were able to process customer transactions faster and provide more efficient service. Customers experienced shorter waiting times during checkout because barcode scanners simplified transaction processing. Employees also benefited because the system reduced repetitive manual tasks and improved workflow organization.

The study further highlighted that barcode technology improved security and inventory control within retail establishments. Because inventory records were updated automatically, businesses were able to reduce the occurrence of missing products and inventory losses caused by recording errors. The researchers explained that barcode-integrated systems also improved warehouse organization and product identification, making inventory-related tasks easier to manage.

Despite the positive outcomes, the study identified several challenges encountered during barcode system implementation. Businesses required financial investment for barcode scanners, software systems, and computerized infrastructure. Employees also needed technical training to understand how to operate barcode equipment effectively. Some organizations initially experienced implementation difficulties because employees were unfamiliar with digital technologies and automated inventory procedures. However, the researchers noted that proper employee training and continuous system usage improved adaptation and operational performance over time.

The study concluded that barcode scanning technologies significantly improve inventory management, transaction efficiency, operational productivity, and customer satisfaction within retail businesses. The researchers emphasized that although implementing barcode systems may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, faster transaction processing, reduced operational delays, and better organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how barcode-integrated inventory systems improve transaction processing, inventory monitoring, and operational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of barcode scanning technology to reduce manual errors, improve sales recording, and enhance customer service.

2.1.15

A foreign study conducted by Chuang, Oliva, and Liao (2006) focused on the role of information technology in supply chain and inventory management within modern business organizations. The study aimed to determine how computerized systems and automation technologies improve inventory coordination, operational efficiency, communication, and organizational productivity. The researchers explained that many organizations continue to encounter inventory-related problems due to manual inventory monitoring procedures and inefficient communication systems. These operational issues often result in delayed inventory updates, inaccurate stock information, poor coordination between departments, and reduced customer satisfaction.

The study emphasized that inventory management is an important component of organizational operations because it directly affects productivity, profitability, and customer service. According to the researchers, businesses that fail to maintain accurate and updated inventory information often experience operational disruptions such as stock shortages, delayed transactions, overstocking, and inefficient inventory planning. Traditional inventory systems rely heavily on paperwork and manual data entry, which consume time and increase the possibility of human error.

To address these operational challenges, the researchers explored how information technology and computerized inventory systems improve business operations and supply chain coordination. The study explained that digital inventory systems centralize inventory information within computerized databases, allowing businesses to access and update inventory records more efficiently. Through automation, organizations are able to monitor inventory movement in real time and improve communication between operational departments.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory visibility and operational coordination. Inventory information could be updated immediately whenever products were sold, transferred, or restocked, allowing businesses to maintain accurate inventory records. The researchers explained that real-time inventory monitoring reduced delays and minimized inventory discrepancies caused by manual recording procedures.

Furthermore, the study found that information technology improved communication and collaboration between departments involved in inventory operations. Purchasing personnel, warehouse staff, sales employees, and management teams were able to access the same inventory information simultaneously, improving organizational coordination and reducing delays in operational processes. The researchers also noted that computerized systems improved workflow management because inventory-related information could be processed and retrieved more efficiently.

Another important finding of the study was the positive effect of automation on decision-making and organizational planning. Managers were able to generate inventory reports more accurately and analyze inventory data in real time. This allowed organizations to identify inventory trends, monitor product movement, and improve purchasing decisions. The study emphasized that accurate inventory information supports better inventory planning and reduces operational losses associated with poor inventory control.

The study also highlighted that computerized inventory systems improved customer service and organizational productivity. Businesses using automated systems were able to provide more accurate information regarding product availability and process customer transactions more efficiently. Customers benefited from faster services and improved product accessibility, while employees experienced reduced workload because repetitive manual inventory tasks were minimized through automation.

Despite the positive findings, the study identified several challenges encountered during the implementation of computerized inventory systems. Organizations required financial investment for technology infrastructure, software systems, and technical support. Employees also needed training programs to understand how to use computerized systems effectively. Some workers initially experienced difficulty adapting to automated technologies because they were more familiar with traditional inventory management methods. However, continuous training and system usage improved employee adaptation and organizational performance over time.

The researchers concluded that information technology and computerized inventory systems significantly improve inventory management, operational coordination, workflow efficiency, and organizational productivity. The study emphasized that automation technologies help businesses reduce operational delays, improve inventory visibility, and strengthen decision-making processes. Although implementation may involve technical and financial challenges, the long-term advantages include improved operational efficiency, better customer service, and enhanced inventory management.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems and real-time monitoring technologies improve inventory management and organizational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of digital inventory systems, centralized databases, and automated inventory updates to reduce manual errors and improve business operations.

2.1.16

A foreign study conducted by Fernie and Sparks (2018) examined retail logistics and inventory management practices in modern retail establishments. The study aimed to determine how automated inventory technologies improve inventory control, operational efficiency, product availability, and customer satisfaction within retail organizations. The researchers explained that retail businesses handle large volumes of products and transactions daily, making inventory management a complex operational task. Businesses relying on manual inventory methods often encounter inventory inaccuracies, stock shortages, delayed product monitoring, and inefficient transaction processing.

The study emphasized that inventory management is essential in maintaining smooth retail operations because inventory directly affects sales performance and customer service. According to the researchers, poor inventory management may result in lost sales opportunities, dissatisfied customers, excessive storage costs, and operational inefficiencies. Traditional inventory systems require employees to manually update stock records and monitor inventory levels, which increases the possibility of human error and delays in inventory processing.

The researchers explored how automated inventory systems improve retail operations by simplifying inventory-related processes and providing real-time inventory information. Through computerized inventory monitoring and barcode-integrated technologies, businesses were able to maintain accurate stock records and update inventory information automatically whenever transactions occurred.

The findings of the study revealed that retail establishments using automated inventory systems experienced significant improvements in inventory accuracy and operational efficiency. Inventory information became more reliable because computerized systems reduced errors associated with manual inventory recording. Businesses were also able to monitor product movement more efficiently and identify low-stock items quickly, reducing the occurrence of product shortages and overstocking.

Furthermore, the study found that automation improved transaction processing and customer service within retail establishments. Barcode scanning technologies simplified checkout procedures and reduced delays during customer transactions. Employees were able to process transactions more efficiently because product information was retrieved automatically from the inventory database. Customers benefited from shorter waiting times and improved service quality.

Another important finding of the study was the improvement of inventory planning and organizational decision-making. Managers were able to access inventory reports more efficiently and analyze product demand using computerized systems. This helped businesses improve purchasing decisions, maintain balanced stock levels, and reduce unnecessary inventory costs. The researchers emphasized that accurate inventory information contributes to better organizational planning and operational productivity.

The study also highlighted that automation improved workflow coordination and warehouse organization. Employees experienced reduced workload because repetitive inventory tasks were simplified through digital systems. Warehouse personnel were also able to locate and organize products more efficiently using barcode technologies and computerized inventory databases.

Despite the positive findings, the study identified several challenges associated with implementing automated inventory systems. Retail businesses required investment in technology infrastructure, barcode equipment, and software systems. Employees also needed technical training to understand how to use the systems properly. Some workers initially experienced difficulty adapting to computerized technologies due to limited technical knowledge and resistance to operational changes. However, the researchers explained that continuous training and organizational support improved employee adaptation over time.

The study concluded that automated inventory systems significantly improve inventory control, operational efficiency, transaction processing, and customer satisfaction within retail establishments. The researchers emphasized that automation technologies help businesses improve workflow management, reduce operational delays, and maintain accurate inventory records. Although implementation may involve technical and financial challenges, the long-term benefits include improved productivity, better inventory management, and enhanced customer service.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems and barcode technologies improve inventory monitoring and retail operations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory updates, barcode scanning systems, and computerized inventory monitoring to improve operational efficiency and customer service.

2.1.17

A foreign study conducted by Waters (2019) examined inventory control systems and their role in improving operational efficiency and inventory management within retail and business organizations. The study aimed to determine how computerized inventory systems contribute to better stock monitoring, inventory planning, and organizational productivity. The researcher explained that inventory management is one of the most important functions within organizations because it directly affects sales performance, customer service, and operational stability. However, many businesses continue to encounter problems related to manual inventory monitoring, including inaccurate stock records, inventory shortages, delayed transactions, and poor inventory coordination.

The study emphasized that traditional inventory management systems are often inefficient because they rely heavily on manual recording, paperwork, and physical inventory checking. These manual procedures consume a significant amount of time and increase the possibility of human error. According to the researcher, businesses handling large volumes of products may experience greater difficulties in maintaining accurate inventory information due to delayed updates and inefficient monitoring systems. As a result, organizations may suffer from inventory discrepancies, overstocking, product expiration, and operational losses.

To address these problems, the study explored the use of computerized inventory control systems in improving inventory-related operations. The researcher explained that automated inventory systems allow organizations to monitor stock levels digitally and update inventory records automatically whenever products are sold, transferred, or restocked. Through centralized databases and computerized monitoring, businesses are able to access accurate inventory information in real time and improve workflow efficiency.

The findings of the study revealed that organizations using computerized inventory systems experienced significant improvements in inventory accuracy and operational productivity. Real-time inventory monitoring reduced inventory discrepancies because inventory records were updated immediately after transactions occurred. Businesses were able to identify low-stock items quickly, improve stock replenishment planning, and reduce losses caused by poor inventory control. The researcher also noted that automation minimized repetitive manual tasks and simplified inventory monitoring procedures.

Furthermore, the study found that computerized inventory systems improved organizational coordination and workflow management. Employees were able to process inventory-related tasks more efficiently because information could be retrieved quickly through digital systems. Managers also benefited from computerized reporting because inventory reports could be generated more accurately and analyzed more efficiently. This improved organizational planning and supported better decision-making regarding inventory purchases and stock management.

Another important finding of the study was the positive effect of automation on customer service. Businesses using automated inventory systems were more capable of maintaining product availability and responding to customer demands efficiently. Customers experienced improved service because products were easier to locate and transactions could be processed faster. The researcher emphasized that customer satisfaction improves when organizations maintain accurate inventory information and reduce delays in operations.

The study also highlighted several technologies commonly integrated into computerized inventory systems, including barcode scanning technologies and digital inventory databases. Barcode systems improved transaction speed and reduced errors associated with manual product encoding. Automated inventory systems also improved inventory visibility by allowing businesses to monitor stock movement continuously and accurately.

Despite the positive findings, the study identified challenges encountered during system implementation. Organizations needed financial resources to acquire computer equipment, software systems, and technical infrastructure. Employees also required technical training to operate computerized inventory systems properly. Some employees initially experienced difficulty adapting to new technologies because they were more familiar with manual inventory methods. However, the researcher explained that continuous training and organizational support improved employee adaptation and system acceptance over time.

The study concluded that computerized inventory control systems significantly improve inventory monitoring, workflow management, organizational productivity, and customer service. The researcher emphasized that although implementing automated inventory systems may involve financial and technical challenges, the long-term benefits include improved inventory accuracy, reduced operational delays, better inventory planning, and enhanced business performance.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems improve inventory accuracy, operational efficiency, and customer service within business organizations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory monitoring, barcode scanning, and computerized inventory control to reduce manual errors and improve inventory-related operations.

2.1.18

A foreign study conducted by Heizer, Render, and Munson (2017) examined the role of inventory management systems in operations management and organizational efficiency. The study aimed to determine how automation technologies improve inventory monitoring, workflow coordination, operational productivity, and customer service within organizations. The researchers explained that inventory management is essential in maintaining efficient business operations because inventory directly affects production, sales, and customer satisfaction. However, businesses relying on manual inventory systems often encounter operational problems such as inaccurate inventory records, delayed transactions, poor inventory coordination, and increased operational costs.

The study emphasized that inventory systems are important because they help organizations maintain adequate stock levels and avoid inventory-related disruptions. According to the researchers, manual inventory methods require employees to perform repetitive tasks such as handwritten inventory recording, manual stock counting, and physical monitoring of products. These procedures consume significant time and increase the possibility of human error, especially in businesses handling large volumes of inventory and transactions daily.

The researchers explored how automated inventory systems improve operational efficiency through computerized inventory monitoring and digital information management. Automated systems allow organizations to update inventory records automatically whenever products are sold, transferred, or restocked. Through centralized inventory databases, employees and managers are able to access accurate inventory information in real time, improving organizational coordination and workflow management.

The findings of the study revealed that organizations using automated inventory systems experienced improved inventory accuracy and operational productivity. Businesses were able to reduce inventory discrepancies caused by manual recording errors and improve the monitoring of stock movement. Real-time inventory updates also enabled organizations to identify low-stock items quickly and improve inventory replenishment planning. As a result, operational delays were reduced and inventory-related processes became more efficient.

Furthermore, the study found that barcode scanning technologies significantly improved transaction speed and inventory management efficiency. Employees using barcode systems were able to process transactions faster because product information could be retrieved automatically from the computerized database. This reduced delays associated with manual encoding and improved customer service within retail and operational environments.

Another important finding of the study was the improvement of organizational planning and decision-making. Managers were able to generate inventory reports more efficiently and analyze inventory movement using computerized systems. This improved purchasing decisions, inventory forecasting, and stock management processes. The researchers explained that accurate inventory information contributes to better operational planning and reduces unnecessary inventory costs.

The study also highlighted that automation improved employee productivity and workflow organization. Because repetitive inventory tasks were minimized, employees were able to focus on more important operational activities. Businesses using computerized inventory systems experienced smoother workflow coordination and improved communication between departments involved in inventory operations.

Despite the positive outcomes, the study identified several challenges related to implementing automated inventory systems. Organizations needed investment in hardware, software, and technical infrastructure necessary for computerized inventory management. Employees also required training programs to understand how to use automated technologies effectively. Some workers initially experienced difficulty adapting to computerized systems because they were accustomed to traditional inventory methods. However, continuous training and operational support improved employee adaptation and system efficiency over time.

The researchers concluded that automated inventory systems significantly improve inventory control, workflow coordination, operational productivity, and customer satisfaction. The study emphasized that although implementing computerized inventory technologies may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, faster transaction processing, reduced operational delays, and enhanced organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems and barcode technologies improve inventory management and organizational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of computerized inventory monitoring, barcode scanning, and real-time inventory updates to improve business operations and customer service.

2.1.19

A foreign study conducted by Jacobs and Chase (2018) explored operations and supply chain management technologies used in retail and manufacturing organizations. The study aimed to determine how computerized inventory systems improve operational efficiency, inventory visibility, workflow coordination, and organizational productivity. The researchers explained that inventory management is a critical operational function because inventory directly affects customer satisfaction, sales performance, and business continuity. Businesses relying on manual inventory methods often experience inventory discrepancies, operational delays, and poor inventory coordination.

The study emphasized that traditional inventory systems are inefficient because they require manual recording, repetitive paperwork, and delayed inventory processing. These manual procedures increase the likelihood of human error and consume significant operational time. According to the researchers, organizations handling large volumes of inventory may struggle to maintain accurate inventory information due to outdated monitoring systems and inefficient operational procedures.

The researchers explored how automation technologies improve inventory operations through computerized monitoring and digital information systems. Automated inventory systems allow organizations to track product movement in real time and update inventory records automatically whenever inventory-related transactions occur. Through centralized databases, employees and management personnel are able to access inventory information more efficiently.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory visibility and operational coordination. Real-time inventory monitoring reduced inventory discrepancies and improved stock management processes. Businesses were able to identify inventory shortages more efficiently and improve stock replenishment planning.

Furthermore, the study found that automation improved workflow efficiency and reduced delays associated with manual inventory monitoring. Employees were able to process transactions and inventory-related tasks more efficiently because information could be accessed quickly through computerized systems. Managers also benefited from digital inventory reporting because inventory data could be analyzed more accurately and efficiently.

Another important finding of the study was the improvement of organizational decision-making and productivity. Businesses using automated systems experienced better operational planning and improved communication between departments involved in inventory operations. The researchers emphasized that accurate inventory information supports better purchasing decisions and improves inventory forecasting.

The study also highlighted that barcode scanning technologies improved transaction speed and reduced errors associated with manual product encoding. Customers benefited from improved service because businesses were able to process transactions faster and maintain accurate product availability information.

Despite the positive findings, the study identified several challenges associated with automation implementation. Organizations required investment in computerized infrastructure and employee training programs. Some workers initially experienced difficulty adapting to automated technologies due to limited technical knowledge and resistance to organizational changes. However, continuous training and operational support improved employee adaptation and system efficiency over time.

The researchers concluded that automated inventory systems significantly improve inventory monitoring, operational efficiency, workflow coordination, and organizational productivity. The study emphasized that automation technologies help organizations reduce operational delays, improve customer service, and strengthen inventory management processes.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems improve inventory visibility and operational efficiency within organizations. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory monitoring, barcode scanning technologies, and computerized inventory databases to improve business operations and inventory management.

2.1.20

A foreign study conducted by Bowersox, Closs, and Cooper (2013) investigated logistics and inventory management technologies used in business organizations and retail establishments. The study aimed to determine how computerized inventory systems and automation technologies improve inventory coordination, operational efficiency, product monitoring, and organizational productivity. The researchers explained that inventory management is an essential operational function because inventory directly affects customer service, workflow efficiency, profitability, and organizational performance. However, businesses that continue to rely on manual inventory systems often encounter inventory discrepancies, delayed product monitoring, operational inefficiencies, and poor coordination between departments.

The study emphasized that traditional inventory management methods are no longer sufficient for modern business environments because organizations handle large volumes of products and transactions daily. According to the researchers, manual inventory systems require employees to perform repetitive inventory recording, physical stock counting, and handwritten reporting processes. These manual procedures consume significant time and increase the likelihood of human error. Inventory inaccuracies may result in stock shortages, delayed transactions, overstocking, product expiration, and customer dissatisfaction.

To improve inventory-related operations, the researchers explored the role of computerized inventory systems and logistics technologies in modern business organizations. The study explained that automated inventory systems centralize inventory information within digital databases, allowing businesses to monitor inventory levels and product movement more accurately. Through real-time inventory updates, organizations are able to maintain accurate inventory records and reduce delays associated with manual inventory monitoring.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory visibility and operational coordination. Automated systems reduced inventory discrepancies because inventory records were updated immediately whenever products were sold, transferred, or restocked. Businesses were also able to identify low-stock products more efficiently and improve inventory replenishment planning.

Furthermore, the study found that automation improved workflow coordination and communication between departments involved in inventory operations. Inventory information could be accessed quickly and simultaneously by warehouse personnel, sales staff, purchasing departments, and managers. This improved operational coordination and reduced delays caused by manual communication and paperwork.

Another important finding of the study was the improvement of customer service and organizational productivity. Businesses using automated inventory systems were able to maintain product availability more effectively and process transactions faster. Customers benefited from improved service because employees could provide accurate product information and complete transactions more efficiently. The researchers emphasized that automation also reduced employee workload by simplifying repetitive inventory tasks and improving workflow organization.

The study further highlighted the importance of barcode scanning technologies and digital inventory databases in improving inventory management efficiency. Barcode systems improved transaction speed and reduced errors associated with manual product encoding. Managers were also able to generate inventory reports more accurately and analyze inventory movement efficiently using computerized systems. This improved organizational planning and supported better inventory-related decision-making.

Despite the positive findings, the study identified several challenges encountered during automation implementation. Organizations needed financial investment to acquire computer equipment, software systems, and technical infrastructure necessary for automation. Employees also required technical training to understand how to use computerized inventory systems effectively. Some workers initially experienced difficulty adapting to automated technologies because they were more familiar with traditional inventory procedures. However, the researchers noted that continuous training and operational support improved employee adaptation and system efficiency over time.

The researchers concluded that computerized inventory systems and logistics technologies significantly improve inventory control, workflow coordination, operational productivity, and customer satisfaction. The study emphasized that although implementing automation technologies may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, reduced operational delays, better workflow management, and enhanced organizational performance.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems and automation technologies improve inventory monitoring and organizational efficiency within retail businesses. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of barcode scanning technologies, real-time inventory updates, and computerized inventory databases to improve inventory management and business operations.

2.1.21

A foreign study conducted by Muller (2011) examined the importance of computerized inventory management systems in retail and warehouse operations. The study aimed to determine how automated inventory systems improve inventory accuracy, stock monitoring, organizational productivity, and operational efficiency within businesses. The researcher explained that inventory management plays an important role in maintaining effective business operations because inventory directly affects sales performance, workflow efficiency, and customer satisfaction. However, organizations relying on manual inventory systems often experience operational problems such as inaccurate inventory records, delayed product monitoring, inventory shortages, and poor inventory coordination.

The study emphasized that manual inventory management methods are inefficient because they require repetitive paperwork, handwritten recording, and manual stock checking procedures. According to the researcher, businesses handling large quantities of inventory often struggle to maintain accurate inventory records due to delayed updates and operational inefficiencies. These inventory problems may result in product shortages, overstocking, unnecessary storage expenses, and customer dissatisfaction.

To address these operational issues, the study explored how computerized inventory systems improve inventory-related operations within organizations. Automated inventory systems allow businesses to monitor inventory movement digitally and update inventory records automatically whenever products are sold, transferred, or restocked. Through centralized digital databases, organizations are able to maintain updated inventory information and improve inventory coordination across departments.

The findings of the study revealed that businesses using computerized inventory systems experienced significant improvements in inventory accuracy and operational efficiency. Inventory discrepancies caused by manual recording errors were minimized because inventory records were updated automatically in real time. Businesses were able to identify low-stock products more efficiently and improve stock replenishment planning.

Furthermore, the study found that automation improved workflow coordination and employee productivity within organizations. Employees were able to process inventory-related tasks more efficiently because information could be accessed quickly through computerized systems. Managers also benefited from digital inventory reports because inventory data could be analyzed more accurately and efficiently. This improved organizational planning and supported better inventory-related decision-making.

Another important finding of the study was the improvement of customer service and transaction efficiency. Businesses using automated inventory systems were able to process transactions faster and maintain accurate product availability information. Customers benefited from improved service because employees could provide accurate information regarding inventory levels and product availability.

The study also highlighted the role of barcode scanning technologies in improving inventory management efficiency. Barcode systems reduced errors associated with manual product encoding and improved transaction speed. The researcher explained that barcode-integrated inventory systems simplified inventory monitoring and improved workflow organization within retail and warehouse environments.

Despite the positive outcomes, the study identified several challenges encountered during automation implementation. Organizations needed financial resources to acquire computer equipment, barcode scanners, and software systems. Employees also required technical training to understand how to use computerized inventory systems effectively. Some workers initially experienced difficulty adapting to automated technologies because they were more familiar with manual inventory methods. However, continuous training and organizational support improved employee adaptation and operational efficiency over time.

The study concluded that computerized inventory systems significantly improve inventory control, workflow management, operational productivity, and customer satisfaction within organizations. The researcher emphasized that although implementing automated inventory systems may involve technical and financial challenges, the long-term benefits include improved inventory accuracy, reduced operational delays, enhanced workflow coordination, and better business performance.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems improve inventory monitoring and organizational efficiency within retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of barcode scanning technologies, real-time inventory updates, and computerized inventory databases to improve inventory-related operations and customer service.

2.1.22

A foreign study conducted by Chopra and Meindl (2016) examined supply chain and inventory management systems used in modern business organizations. The study aimed to determine how automation technologies improve inventory visibility, workflow coordination, inventory planning, and organizational productivity. The researchers explained that inventory management is a critical business function because inventory directly affects customer service, operational efficiency, and organizational profitability. Businesses relying on traditional inventory systems often experience delayed inventory updates, inaccurate stock information, inventory shortages, and poor operational coordination.

The study emphasized that modern business environments require accurate and real-time inventory information to maintain efficient operations and respond to customer demands effectively. According to the researchers, manual inventory methods are inefficient because they rely heavily on paperwork, physical inventory checking, and delayed inventory reporting. These manual procedures increase operational costs and the possibility of human error.

The researchers explored how computerized inventory systems improve inventory management through digital monitoring and real-time inventory updates. Automated inventory systems allow organizations to monitor product movement continuously and update inventory records automatically whenever transactions occur. Through centralized databases and computerized technologies, organizations are able to improve workflow coordination and inventory planning.

The findings of the study revealed that businesses using automated inventory systems experienced significant improvements in inventory visibility and operational efficiency. Real-time inventory updates reduced inventory discrepancies and improved inventory monitoring processes. Businesses were able to identify low-stock products quickly and improve stock replenishment planning.

Furthermore, the study found that automation improved workflow coordination and organizational productivity. Employees were able to process inventory-related operations more efficiently because information could be accessed quickly through computerized systems. Managers also benefited from digital inventory reports because they could analyze inventory movement and improve inventory-related decision-making processes.

Another important finding of the study was the improvement of customer service and transaction efficiency. Businesses using automated inventory systems were able to maintain product availability and process customer transactions more efficiently. Customers experienced improved service because businesses could provide accurate product information and reduce delays in operations.

The study also highlighted the role of barcode scanning technologies and digital databases in improving inventory management efficiency. Barcode systems reduced manual encoding errors and improved transaction speed. Managers were also able to access inventory information in real time and improve inventory planning.

Despite the positive findings, the study identified several implementation challenges such as financial costs, technical infrastructure requirements, and employee adaptation to automated systems. However, the researchers emphasized that the long-term benefits of automation outweigh the implementation difficulties because automation improves operational efficiency and organizational productivity.

The study concluded that computerized inventory systems significantly improve inventory visibility, operational coordination, organizational productivity, and customer service. The researchers emphasized that automation technologies are important in modern inventory and supply chain management because they simplify operational processes and improve inventory-related decision-making.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems improve inventory management and organizational performance. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory updates, barcode technologies, and computerized inventory monitoring to improve operational efficiency and customer service.

2.1.23

A foreign study conducted by Silver, Pyke, and Peterson (1998) explored inventory management and production planning systems within business organizations. The study aimed to determine how computerized inventory systems improve inventory planning, operational efficiency, workflow coordination, and inventory accuracy. The researchers explained that inventory management is important because inventory directly affects production, sales operations, and customer satisfaction. Businesses relying on manual inventory systems often encounter inventory discrepancies, delayed transactions, and inefficient operational procedures.

The study emphasized that traditional inventory methods are inefficient because they rely on manual inventory checking and handwritten records. These manual procedures increase operational delays and the likelihood of human error. Organizations handling large inventory volumes may struggle to maintain accurate inventory records because information is not updated immediately during transactions.

The researchers explored how computerized inventory systems improve inventory-related operations through automation and digital monitoring technologies. Automated systems allow organizations to update inventory records in real time whenever products are sold, transferred, or restocked. This improves inventory visibility and reduces operational delays associated with manual inventory procedures.

The findings of the study revealed that businesses using computerized inventory systems experienced improved inventory accuracy and workflow coordination. Real-time inventory updates minimized inventory discrepancies and improved stock monitoring processes. Businesses were able to identify low-stock items quickly and improve inventory replenishment planning.

Furthermore, the study found that automation improved organizational productivity and operational planning. Employees were able to process inventory-related tasks more efficiently because inventory information could be accessed quickly through digital systems. Managers also benefited from computerized inventory reports because inventory movement could be analyzed more accurately.

Another important finding of the study was the improvement of customer service and transaction processing. Businesses using automated inventory systems were able to maintain accurate product availability information and reduce delays during transactions. Customers experienced improved service because businesses could process transactions more efficiently.

Despite the positive outcomes, the study identified several challenges associated with implementing computerized inventory systems. Organizations needed financial resources to acquire computerized technologies and technical infrastructure. Employees also required training programs to understand how to use digital inventory systems effectively.

The study concluded that computerized inventory systems significantly improve inventory monitoring, workflow coordination, organizational productivity, and customer service. The researchers emphasized that automation technologies reduce operational inefficiencies and improve inventory-related decision-making processes.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems improve inventory accuracy and operational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the integration of real-time inventory monitoring and computerized inventory databases to improve inventory management processes.

2.1.24

A foreign study conducted by Christopher (2016) investigated logistics and supply chain management practices in modern retail organizations. The study aimed to determine how automated inventory technologies improve inventory visibility, operational efficiency, workflow coordination, and customer service. The researcher explained that inventory management is an important operational function because inventory directly affects organizational productivity and customer satisfaction. Businesses using manual inventory systems often experience delayed inventory updates, stock shortages, operational inefficiencies, and poor inventory coordination.

The study emphasized that modern retail businesses require accurate inventory information and efficient operational processes to remain competitive. Traditional inventory systems rely heavily on manual inventory monitoring and paperwork, which increase operational delays and human error. According to the researcher, businesses handling large volumes of products require automated systems to improve inventory coordination and operational efficiency.

The findings of the study revealed that businesses using automated inventory systems experienced improved inventory visibility and workflow management. Real-time inventory updates reduced inventory discrepancies and improved product monitoring. Businesses were able to identify inventory shortages quickly and improve stock replenishment planning.

Furthermore, the study found that automation improved customer service because businesses were able to maintain accurate product availability information and process transactions more efficiently. Employees also benefited because automated systems simplified repetitive inventory tasks and improved workflow coordination.

Another important finding of the study was the improvement of organizational productivity and inventory planning. Managers were able to access inventory reports more accurately and improve inventory-related decision-making processes through computerized systems.

Despite the positive findings, the study identified several implementation challenges such as financial investment, technical infrastructure requirements, and employee adaptation to computerized technologies. However, the researcher emphasized that the long-term benefits of automation outweigh these challenges because automation improves operational efficiency and organizational performance.

The study concluded that automated inventory systems significantly improve inventory management, workflow efficiency, and customer service within retail organizations. The researcher emphasized that automation technologies are essential in improving inventory operations and organizational productivity.

This foreign study is highly relevant to the present study because it demonstrates how automated inventory systems improve inventory visibility and operational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of computerized inventory monitoring and real-time inventory updates to improve business operations and customer service.

2.1.25

A foreign study conducted by Nahmias and Olsen (2015) examined production and inventory management systems used in modern business organizations. The study aimed to determine how computerized inventory systems improve inventory planning, operational efficiency, inventory coordination, and organizational productivity. The researchers explained that inventory management is essential because inventory directly affects production operations, customer satisfaction, and organizational performance. Businesses relying on traditional inventory methods often encounter operational problems such as delayed inventory updates, inaccurate stock records, and inefficient workflow coordination.

The study emphasized that manual inventory systems are inefficient because they rely on repetitive paperwork, handwritten recording, and physical inventory checking procedures. These manual methods increase operational delays and the possibility of human error. According to the researchers, organizations handling large inventory volumes require computerized systems to maintain accurate inventory information and improve workflow efficiency.

The findings of the study revealed that businesses using computerized inventory systems experienced improved inventory visibility and operational productivity. Real-time inventory updates minimized inventory discrepancies and improved inventory coordination. Businesses were able to monitor stock movement more efficiently and improve inventory replenishment planning.

Furthermore, the study found that automation improved organizational decision-making because managers could access inventory information and reports more accurately through computerized systems. Businesses also experienced improved workflow coordination because employees could process inventory-related tasks more efficiently.

Another important finding of the study was the improvement of customer service and operational performance. Businesses using automated inventory systems were able to maintain accurate product availability information and reduce delays during transactions. Customers experienced improved service because organizations could respond to customer demands more effectively.

The study also highlighted the importance of digital inventory databases and automation technologies in improving organizational efficiency. Automated systems reduced repetitive manual tasks and simplified inventory-related operations within organizations.

Despite the positive findings, the study identified several challenges related to system implementation such as financial investment, technical infrastructure requirements, and employee adaptation to computerized technologies. However, the researchers emphasized that the long-term advantages of automation outweigh the challenges because automation improves operational efficiency and inventory management performance.

The study concluded that computerized inventory systems significantly improve inventory planning, workflow coordination, organizational productivity, and customer satisfaction. The researchers emphasized that automation technologies are important in improving inventory operations and business performance.

This foreign study is highly relevant to the present study because it demonstrates how computerized inventory systems improve inventory monitoring and operational efficiency. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of real-time inventory updates, digital inventory databases, and automated inventory monitoring to improve inventory management and organizational productivity.

2.2 Local Studies

Local studies also support the use of automated inventory systems and web-based Point of Sale technologies in improving retail and business operations in the Philippines. These studies demonstrate that digital systems help improve inventory accuracy, transaction speed, record management, and operational efficiency.

2.2.1

A local study conducted by Dela Cruz and Ramos (2018) focused on the development of a Point of Sale (POS) and Inventory Management System for small retail businesses in the Philippines. The study aimed to address operational problems associated with manual inventory monitoring, handwritten sales recording, and inefficient transaction processing commonly experienced by small and medium-sized enterprises. According to the researchers, many local businesses still relied on traditional inventory management practices that involved manual computation, physical inventory checking, and paper-based transaction recording. These methods often resulted in inaccurate inventory records, delayed transactions, inventory discrepancies, and operational inefficiencies.

The researchers emphasized that inventory management is one of the most important operational functions in retail businesses because inventory directly affects sales performance, customer service, and organizational productivity. However, businesses using manual inventory systems often experience difficulties in maintaining accurate stock records because inventory information is not updated immediately after sales transactions. As a result, businesses may encounter stock shortages, overstocking, product expiration, and loss of sales opportunities.

To address these operational challenges, the researchers developed a computerized Point of Sale and Inventory Management System designed to automate sales recording and inventory monitoring processes. The proposed system allowed businesses to process transactions digitally while automatically updating inventory levels whenever products were sold. Through a centralized database, inventory information could be accessed quickly and updated in real time.

The findings of the study revealed that the implementation of the automated system significantly improved inventory accuracy and operational efficiency within small retail businesses. Inventory discrepancies caused by manual recording errors were minimized because inventory data was updated automatically during transactions. Businesses were also able to monitor stock movement more effectively and identify low-stock products immediately.

Furthermore, the study found that the automated system improved transaction speed and customer service. Employees no longer needed to encode product information manually because the POS system processed sales transactions automatically. Customers experienced shorter waiting times during checkout, while employees benefited from reduced workload and improved workflow organization.

Another important finding of the study was the improvement of organizational planning and inventory-related decision-making. Managers were able to generate inventory and sales reports more efficiently using computerized systems. This helped businesses analyze product demand, monitor inventory trends, and improve purchasing decisions.

Despite the positive findings, the study identified several challenges encountered during system implementation. Businesses needed financial resources to acquire computer equipment and install the software system. Employees also required technical training to understand how to operate the automated system properly. Some workers initially experienced difficulty adapting to computerized technologies because they were more familiar with traditional inventory procedures.

The researchers concluded that computerized POS and inventory systems significantly improve inventory management, transaction processing, operational productivity, and customer service within retail businesses. The study emphasized that automation technologies help reduce manual errors and improve workflow efficiency.

This local study is highly relevant to the present study because it demonstrates how automated inventory systems improve inventory monitoring and sales recording within Philippine retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of computerized inventory monitoring, real-time inventory updates, and automated transaction processing to improve operational efficiency and customer service.

2.2.2

A local study conducted by Santos, Garcia, and Lopez (2020) examined the implementation of a web-based inventory monitoring system for retail stores in the Philippines. The study aimed to determine how computerized inventory systems improve inventory tracking, stock management, operational efficiency, and business organization within local retail establishments. The researchers explained that many Philippine retail businesses still experience operational difficulties because they rely heavily on manual inventory management and traditional sales recording procedures.

According to the study, manual inventory systems are inefficient because inventory information is often recorded using paper-based methods and handwritten reports. These traditional procedures increase the likelihood of human error and delayed inventory updates. Businesses may experience inaccurate stock records, misplaced products, delayed inventory checking, and inventory shortages. The researchers also noted that poor inventory monitoring negatively affects customer service because businesses may fail to maintain adequate stock levels.

To address these operational issues, the researchers developed a web-based inventory monitoring system that allowed businesses to monitor inventory levels digitally and access inventory information in real time. The proposed system centralized inventory data within an online database, enabling employees and managers to access inventory records more efficiently.

The findings of the study revealed that the web-based inventory monitoring system significantly improved inventory visibility and operational efficiency. Businesses were able to update inventory records automatically whenever products were sold or restocked. This reduced inventory discrepancies caused by delayed manual recording and improved the accuracy of stock information.

Furthermore, the study found that real-time inventory monitoring improved inventory planning and organizational coordination. Managers were able to identify low-stock items quickly and improve purchasing decisions using computerized inventory reports. Employees also benefited because the system reduced repetitive inventory tasks and improved workflow organization.

Another important finding of the study was the improvement of customer satisfaction and transaction efficiency. Businesses using the automated inventory system were able to maintain product availability more effectively and process transactions faster. Customers experienced improved service because inventory information could be accessed quickly and accurately.

The study also highlighted the importance of digital technologies in improving operational productivity within Philippine retail establishments. The researchers explained that web-based systems provide businesses with centralized inventory information and simplify inventory-related operations.

Despite the positive outcomes, the study identified several challenges encountered during system implementation. Businesses required internet connectivity, computer equipment, and technical infrastructure to operate the web-based system effectively. Employees also needed training programs to understand how to use the system properly.

The researchers concluded that web-based inventory monitoring systems significantly improve inventory management, workflow efficiency, customer service, and operational productivity within retail businesses. The study emphasized that automation technologies help reduce manual errors and improve inventory-related decision-making processes.

This local study is highly relevant to the present study because it demonstrates how web-based inventory systems improve inventory monitoring and operational efficiency within Philippine retail establishments. Similar to the proposed Automated Inventory Management System for pharmacies and convenience stores, the study supports the use of computerized inventory databases, real-time inventory monitoring, and digital inventory management technologies.

2.2.2

A local study conducted by Villanueva and Perez (2021) focused on the development of a barcode-based Point of Sale system for convenience stores in the Philippines. The study aimed to determine how barcode scanning technologies improve transaction processing, inventory accuracy, customer service, and operational productivity within convenience store operations. The researchers explained that many local convenience stores continue to use traditional transaction methods and manual inventory systems, which often result in delayed transactions, inaccurate inventory records, and poor workflow coordination.

The study emphasized that barcode scanning technology has become an important component of modern inventory and retail management systems because it improves the speed and accuracy of transaction processing. According to the researchers, businesses using manual encoding procedures often encounter operational delays because employees need to enter product information manually during transactions. This process increases the likelihood of human error and negatively affects customer satisfaction.

To address these operational issues, the researchers developed a barcode-integrated Point of Sale system capable of processing transactions automatically while updating inventory records in real time. The proposed system used barcode scanners to retrieve product information from a computerized database whenever products were scanned during transactions.

The findings of the study revealed that barcode scanning technologies significantly improved transaction speed and inventory accuracy within convenience stores. Employees were able to process transactions faster because product information was retrieved automatically through barcode scanning. This minimized manual encoding errors and improved inventory visibility.

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2.1.1

Another related foreign study is Design and Implementation of Computerized Stock Management Systems: A Case Study of MIDE Supermarket by Akinsanya (2019). The study developed a computerized stock management system to improve inventory monitoring and reduce the problems caused by manual stock recording. The researchers explained that traditional inventory methods are time-consuming and prone to errors, which may affect business performance. The proposed system provided automated inventory tracking and organized record management for supermarkets. This study is relevant to the present study because both studies focus on improving inventory monitoring through digital systems. Similar to the proposed system, the study aimed to reduce manual work and improve inventory accuracy.

2.1.2

Another foreign study related to the present project is Design of Computerized Inventory Management Systems for Supermarkets by Abisoye, Fatoba, and Blessing (2013). The study developed a computerized inventory system that automated product monitoring and sales recording for supermarkets. The researchers found that the system improved efficiency, minimized stock management problems, and reduced the risks associated with manual inventory processes. This study supports the present project because it demonstrates how automated inventory systems can improve operational efficiency and record management in retail establishments. However, the present study specifically focuses on pharmacies and convenience stores and includes barcode scanning and real-time stock updates.

2.1.3

Another related foreign study is Inventory Management Automation and Performance of Supermarkets in Western Kenya by Samuel and Ondiek (2014). The study investigated how automation technologies affect supermarket performance and inventory management efficiency. The findings showed that automated inventory systems improved inventory control, reduced product shortages, and enhanced business operations. This is relevant to the present study because it confirms that automation can improve inventory accuracy and operational performance in retail businesses.

2.2 Local Studies

Local studies also support the use of automated inventory systems and web-based Point of Sale technologies in improving retail and business operations in the Philippines. These studies demonstrate that digital systems help improve inventory accuracy, transaction speed, record management, and operational efficiency.

One related local study is Point of Sale and Inventory Management System for Small Businesses by Dela Cruz and Ramos (2018). The study developed a POS and inventory management system designed to help small businesses improve transaction processing and inventory monitoring. The researchers explained that manual recording often results in inventory discrepancies and slow transaction processing. The developed system automated sales recording and inventory updates, improving efficiency and reducing human errors. This study is relevant to the present project because both systems focus on automating sales and inventory processes to improve business operations.

2.2.1

Another related local study is Web-Based Inventory Monitoring System for Retail Stores by Santos, Garcia, and Lopez (2020). The study developed a web-based inventory monitoring system that allowed store owners and staff to track inventory levels and product movement in real time. The study found that digital inventory systems improve organization, minimize stock discrepancies, and provide faster access to inventory information. This study supports the present project because both studies aim to replace manual inventory monitoring with a more efficient and organized digital solution.

2.2.3

Another local study related to the present project is Barcode-Based Point of Sale System for Convenience Stores by Villanueva and Perez (2021). The study developed a barcode-integrated POS system to improve checkout speed and inventory accuracy in convenience stores. The researchers found that barcode scanning reduced transaction delays and improved sales recording efficiency. This study is relevant because the proposed Automated Inventory Management System also integrates barcode scanning technology to reduce transaction time and improve operational performance.

Summary and Comparison of Local Studies

The reviewed local studies show that automated inventory and Point of Sale systems are useful in improving retail operations in the Philippine setting. These systems help improve inventory organization, transaction speed, sales recording, and inventory monitoring.

However, the reviewed local studies generally focus on general inventory management or POS operations for retail businesses. In contrast, the present study specifically focuses on pharmacies and convenience stores and integrates features such as automated inventory tracking, barcode scanning, real-time stock updates, and low stock alerts. The proposed system is also intended to improve both inventory management and operational efficiency within the selected business environment.

2.3 Related Literature

Information Systems

An information system is a combination of technologies, processes, and data used to collect, process, store, and distribute information within an organization. Information systems are important in business operations because they improve record management, data accuracy, and accessibility. In inventory and retail management, information systems help replace manual processes with automated digital operations. According to Saleem (2020), automated inventory systems improve inventory monitoring and operational efficiency by reducing manual work and improving real-time access to inventory information. This supports the present study because the proposed Automated Inventory Management System also aims to improve inventory monitoring and sales recording through digital automation.

Inventory Management Systems

Inventory management systems are computerized solutions designed to monitor product quantities, stock movement, and inventory levels within a business establishment. These systems help businesses maintain accurate inventory records, reduce stock discrepancies, and improve operational efficiency.

According to Akinsanya (2019), computerized inventory systems improve stock monitoring and reduce errors commonly associated with manual inventory management. This supports the present study because the proposed system aims to automate inventory tracking and provide real-time inventory updates for pharmacies and convenience stores.

Point of Sale (POS) Systems

A Point of Sale (POS) system is a computerized system used to process sales transactions and record product purchases. POS systems are widely used in retail businesses because they improve transaction speed, automate sales recording, and support inventory management.

The integration of POS systems with inventory management allows businesses to automatically update inventory levels whenever products are sold. This helps reduce manual encoding and improves transaction accuracy. The present study applies this concept by integrating a POS feature into the Automated Inventory Management System.

Barcode Scanning Technology

Barcode scanning technology is commonly used in inventory and retail systems to identify products quickly and accurately. Barcode scanners help improve transaction speed, reduce manual encoding, and minimize human error.

According to previous studies on barcode-integrated systems, barcode scanning significantly improves checkout efficiency and inventory accuracy. This supports the present study because the proposed system includes barcode scanning functionality to improve transaction processing in pharmacies and convenience stores.

Software Development

Software development methodology is important because it provides a structured process for designing, developing, testing, and improving software systems.

The present study uses the Agile Development Model because it supports flexibility, continuous improvement, and adaptation based on user feedback. According to Ríos and Pedreira-Souto (2019), agile methodologies are effective in developing web-based systems because they allow developers to refine system features continuously and respond to changing requirements. This supports the present study because the researchers may improve the proposed system based on testing and user evaluation.

Similarly, Qumer and Henderson-Sellers (2008) explained that agile methods help development teams improve software quality and system functionality while responding to user needs effectively. This is relevant to the current study because the proposed Automated Inventory Management System may require continuous improvements during development and testing.

Technologies Used

The proposed system is a web-based application that uses technologies for front-end interface design, server-side processing, and database management.

Web-based systems are commonly used in inventory and retail operations

2.2 Local Studies

The study titled “Inventory Management System of the Selected Pharmacies in the City of Baliwag: An Assessment” focused on evaluating how computerized inventory systems help pharmacies manage their daily operations. The researchers explained that inventory management is an essential part of a business because it allows owners and staff to properly monitor the availability of products, maintain accurate stock records, and avoid shortages or overstocking of medicines and supplies. Since pharmacies handle many different products every day, the use of a manual inventory process may lead to errors, delays, and difficulty in tracking stocks.

The study revealed that implementing a computerized inventory management system improved the efficiency of pharmacy operations. Through automation, employees were able to record and monitor inventory faster and with greater accuracy compared to manual methods. The system also helped reduce human errors in recording product quantities, expiration dates, and stock availability. In addition, the researchers emphasized that computerized systems provide organized and accessible records, making it easier for businesses to track transactions and generate inventory reports whenever needed.

Furthermore, the study highlighted that inventory management systems contribute to better customer service because employees can quickly check product availability and process transactions more efficiently. The use of technology also helped pharmacies save time and improve productivity in their daily operations. According to the findings of the study, businesses that adopt computerized systems are more capable of maintaining accurate inventory data and improving overall operational performance.

This local study is relevant to the present research because it demonstrates the importance of using a computerized inventory management system in improving business operations and inventory control. Similar to the present study, it supports the idea that automation can help businesses reduce manual errors, organize records efficiently, and improve the monitoring of stocks and supplies. The findings of the study also serve as a guide in developing a system that is reliable, accurate, and beneficial for both business owners and employees.

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The study revealed that implementing a computerized inventory management system improved the efficiency of pharmacy operations. Through automation, employees were able to record and monitor inventory faster and with greater accuracy compared to manual methods. The system also helped reduce human errors in recording product quantities, expiration dates, and stock availability. In addition, the researchers emphasized that computerized systems provide organized and accessible records, making it easier for businesses to track transactions and generate inventory reports whenever needed.

Furthermore, the study highlighted that inventory management systems contribute to better customer service because employees can quickly check product availability and process transactions more efficiently. The use of technology also helped pharmacies save time and improve productivity in their daily operations. According to the findings of the study, businesses that adopt computerized systems are more capable of maintaining accurate inventory data and improving overall operational performance.

This local study is relevant to the present research because it demonstrates the importance of using a computerized inventory management system in improving business operations and inventory control. Similar to the present study, it supports the idea that automation can help businesses reduce manual errors, organize records efficiently, and improve the monitoring of stocks and supplies. The findings of the study also serve as a guide in developing a system that is reliable, accurate, and beneficial for both business owners and employees.

2.3 Related Literature

Silberschatz, Korth, and Sudarshan (2019) established that well-designed database systems are fundamental to ensuring accurate and efficient data handling in information systems. In the context of inventory management, their work supports the use of relational database structures that allow for real-time querying, transactional integrity, and scalable data storage — all of which are critical components of an Automated Inventory Management System (AIMS).

Sharp, Rogers, and Preece (2019) emphasized that user-centered design is a foundational principle in developing efficient and usable software systems. Their framework advocates for continuous user involvement throughout the design and development process to ensure that systems align with actual user needs. This principle is directly applicable to the development of a pharmacy and convenience store management system, where ease of use by non-technical staff is essential.

Pressman and Maxim (2019) further provided a comprehensive framework for software engineering practices, including requirements analysis, system modeling, and quality assurance testing. Their practitioner-oriented approach supports the development methodology applied in this study, particularly in the structured design and iterative testing phases of the proposed AIMS.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies demonstrate that automated inventory systems significantly improve operational efficiency and accuracy. However, limited research focuses on integrated systems for pharmacy and convenience store operations. This study addresses this gap by developing an Automated Inventory Management System, aligning with established MIS frameworks and proven system implementations.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The "Automated Inventory Management System for Pharmacy and Convenience Store" is designed to address the challenges of manual inventory tracking and sales management in pharmacies and convenience stores. This research aims to develop a comprehensive local network system that operates offline and streamlines operations including product management, sales processing, and inventory monitoring for both pharmacy and convenience store items.

3.2 System Development Methodology

The Agile Development Model will be used, consisting of the following phases:

1. Requirements Analysis
 2. System Design
 3. Development
 4. Testing
 5. Deployment (Prototype)
-

3.3 System Architecture

The system will follow a client-server architecture running on a local network, where users interact through a local interface, and the server manages data processing and inventory management. The system operates offline within the local network.

3.4 Tools and Technologies Used

- **Frontend:** HTML, CSS, JavaScript
 - **Backend:** Python Flask
 - **Database:** MySQL
 - **DevelopmentTools:** Visual Studio Code
-

3.5 Data Gathering Techniques

We conduct interviews with pharmacy staff and customers to collect firsthand information about their experiences, opinions, and needs. Interviews allow us to gather detailed and personal insights directly from the people who work in or use the pharmacy. This helps us understand challenges, expectations, and areas for improvement.

3.6 Testing Procedures

- . Unit Testing
 - Tested individual Python functions
 - Verified database CRUD operations
 - Checked JavaScript functions
 - . System Testing
 - Tested complete user workflows
 - Verified admin and cashier operations
 - Tested sales processing end-to-end
 - Validated inventory updates
 - . User Acceptance Testing
 - Admin tested all management features
 - Cashiers tested sales processing
 - Validated system meets requirements
-

3.7 Evaluation Criteria

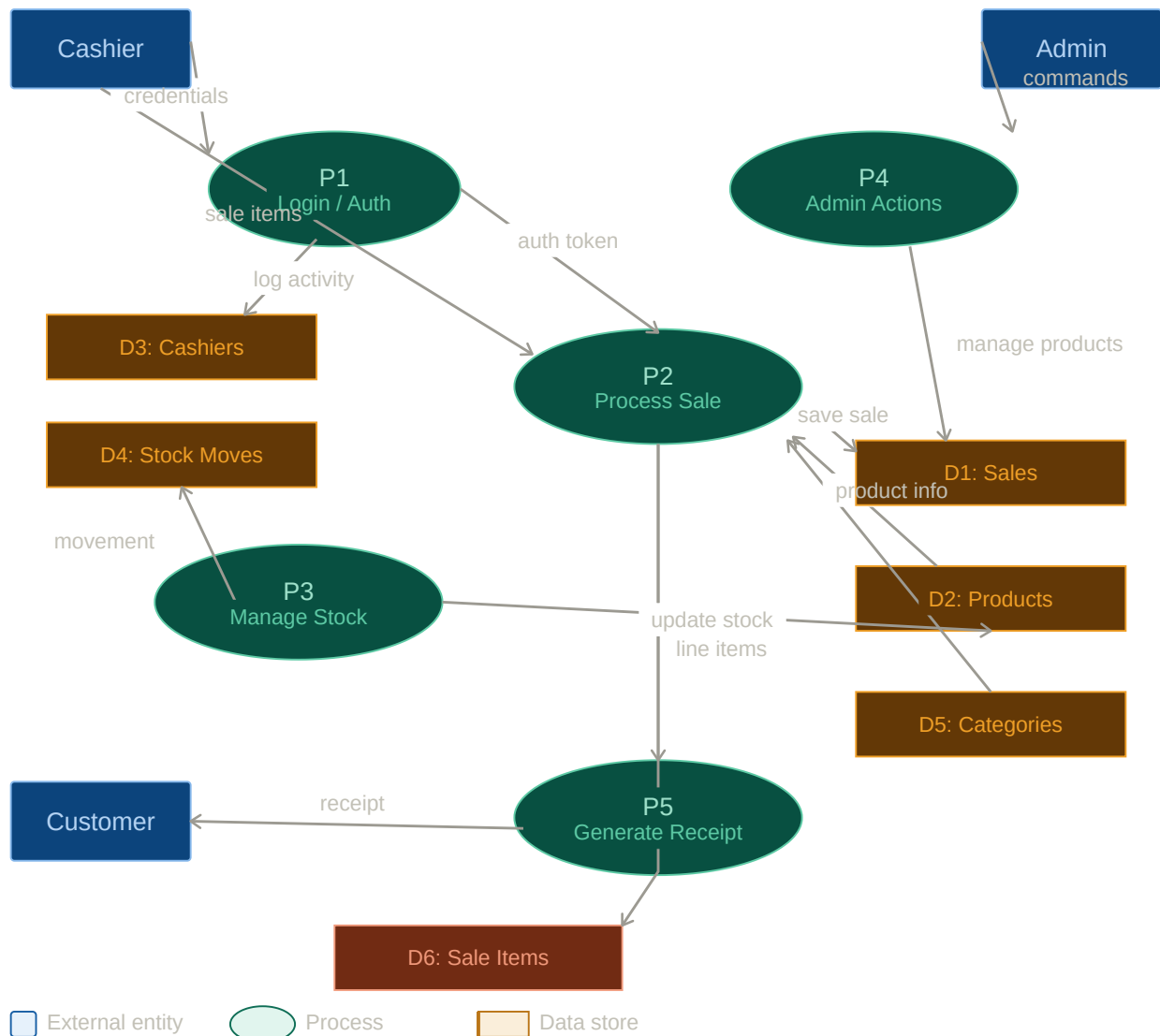
The system will be evaluated based on the following criteria:

- Functionality
 - Usability
 - Accuracy of Machine Learning Predictions
 - Security
 - Efficiency
-

SUPPORTING DIAGRAMS (Included within Chapter 3)

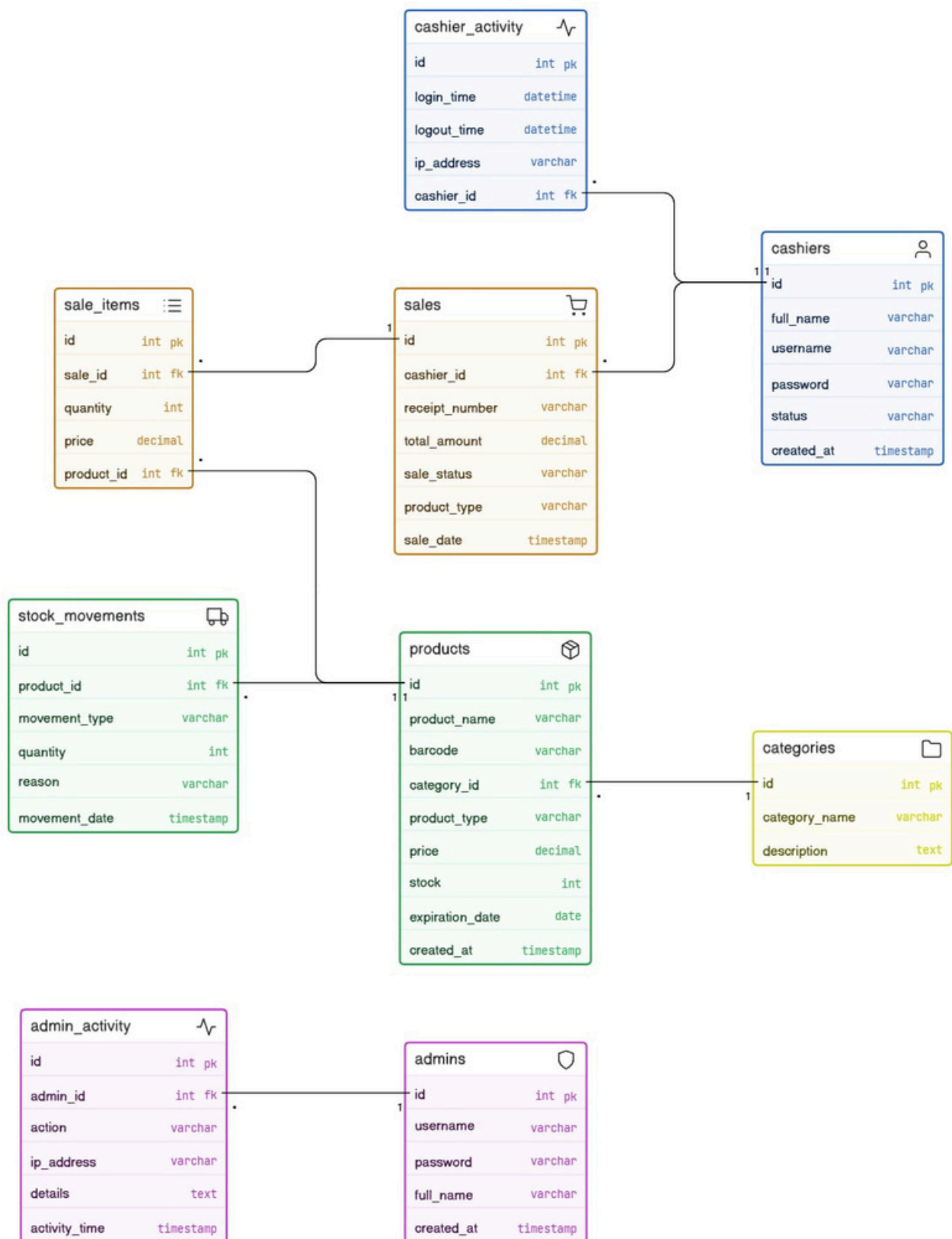
Data Flow Diagram (DFD – Level 0)

User -> PharmaCon Application -> Database



Entity Relationship Diagram (ERD)

Entities:



Use Case Diagram

Actors:

- . Administrator
- . Cashier

Use Cases:

- . Login/Logout (Admin, Cashier)
 - . Manage Products (Admin)
 - . Manage Categories (Admin)
 - . View Inventory (Admin, Cashier)
 - . Process Sale (Cashier)
 - . View Sales History (Admin, Cashier)
 - . Manage Cashiers (Admin)
 - . View Activity Logs (Admin)
 - . Monitor Stock Levels (Admin)
 - . Process Expiring Products (Admin)
-

End of Proposal

